

PARTH ARORA

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Summary

Computer Engineering undergraduate (CGPA: 9.18) specializing in Machine Learning and Generative AI. Experienced in building Retrieval-Augmented Generation (RAG) applications using Python, LangChain, and open-source LLMs. Familiar with developing REST APIs using FastAPI. Solved 150+ Data Structures and Algorithms problems on LeetCode. Seeking a Software Engineering or AI/ML internship or full-time role.

Education

Thapar Institute of Engineering & Technology (TIET)

Bachelor of Engineering (B.E.) in Computer Engineering — CGPA: 9.18 / 10

2024 – 2028

Punjab, India

Technical Skills

Programming Languages: C++, Python, JavaScript

Backend Development: FastAPI, REST API Design, API Integration

AI / Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), LangChain, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering

Databases: SQL, MongoDB, Firebase, ChromaDB (Vector Database)

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Computer Networks

Tools & Platforms: Git, GitHub, Docker, VS Code, Postman

Projects

EchoMind – Local AI Video & Meeting Assistant

GitHub Repository

Python, Whisper, LangChain, Llama 3, ChromaDB

- Built a privacy-first AI meeting assistant that processes meeting recordings locally without relying on cloud APIs.
- Reduced audio transcription time by 96% (from 54 minutes to ~2 minutes for a 22-minute video) by integrating faster-whisper (CTranslate2 INT8 quantization) and optimizing audio preprocessing.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using LangChain, HuggingFace sentence embeddings, and ChromaDB for semantic transcript search and conversational question answering.
- Developed timestamp-aware transcript retrieval, enabling users to ask chronological questions about long meetings and lectures.
- Designed a modular pipeline for transcription, semantic retrieval, and question answering to support efficient local AI workflows.

NeuroLearn – Neuro-Adaptive Gamified Learning Platform

GitHub Repository

Python, LangChain, LSTM, React, Machine Learning

- Designed a multi-agent adaptive learning platform delivering personalized tutoring, multilingual instruction, and AI-generated educational content.
- Implemented LSTM-based knowledge tracing and spaced-repetition algorithms to drive adaptive review scheduling and difficulty personalization.
- Built ADHD-aware feedback logic by integrating eye-tracking attention data with a React-based mastery dashboard, improving learner engagement tracking.
- Designed modular backend components supporting personalized learning workflows and recommendation pipelines.

Problem Solving

Solved 150+ problems on LeetCode spanning Graphs, Dynamic Programming, Greedy Algorithms, Binary Search, and advanced Data Structures.

Relevant Coursework

Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

Achievements

3rd Place – MLSC Hackathon (NeuroLearn) | **Winner** – CTD Ideathon | **3rd Place** – OpenMic 3.0